

Exercise 3

Experiment 1: Static Variable

Create a class **Employee** with the following:

- Employee ID (instance variable)
- Employee Name (instance variable)
- Company Name (static variable)

Create 3 employee objects and display the details. Change the company name once using the class name and show that the change is reflected for all employees.

```
class Employee {
    // Instance variables
    int employeeId;
    String employeeName;
    // Static variable (shared by all objects)
    static String companyName = "ABC Technologies";
    // Constructor
    Employee(int id, String name) {
        employeeId = id;
        employeeName = name;
    }

    // Method to display employee details
    void display() {
        System.out.println("Employee ID : " + employeeId);
        System.out.println("Employee Name : " + employeeName);
        System.out.println("Company Name : " + companyName);
        System.out.println();
    }
}

public class Main {
    public static void main(String[] args) {

        // Creating 3 employee objects
        Employee emp1 = new Employee(101, "John");
        Employee emp2 = new Employee(102, "Alice");
        Employee emp3 = new Employee(103, "David");

        System.out.println("Employee Details Before Changing Company Name");
        System.out.println("-----");
        emp1.display();
        emp2.display();
        emp3.display();

        // Changing company name using class name
        Employee.companyName = "XYZ Solutions";

        System.out.println("Employee Details After Changing Company Name");
        System.out.println("-----");
```

```
emp1.display();  
emp2.display();  
emp3.display();  
}  
}
```

Employee Details Before Changing Company Name

Employee ID : 101
Employee Name : John
Company Name : ABC Technologies

Employee ID : 102
Employee Name : Alice
Company Name : ABC Technologies

Employee ID : 103
Employee Name : David
Company Name : ABC Technologies

Employee Details After Changing Company Name

Employee ID : 101
Employee Name : John
Company Name : XYZ Solutions

Employee ID : 102
Employee Name : Alice
Company Name : XYZ Solutions

Employee ID : 103
Employee Name : David
Company Name : XYZ Solutions

Experiment 2: Visitor Counter

Create a class **VisitorCounter** with:

- A static variable count
- A constructor that increments count

Create multiple objects and display the total number of visitors.

```
class VisitorCounter {  
  
    static int count = 0;  
    VisitorCounter() {  
        count++;  
    }  
    static void displayCount() {  
        System.out.println("Total Visitors = " + count);  
    }  
}
```

```

public class Main {
    public static void main(String[] args) {
        VisitorCounter v1 = new VisitorCounter();
        VisitorCounter v2 = new VisitorCounter();
        VisitorCounter v3 = new VisitorCounter();
        VisitorCounter v4 = new VisitorCounter();

        VisitorCounter.displayCount();
    }
}

```

Total Visitors = 4

Experiment 3: Static Method

Create a class **Calculator** containing static methods to:

- Find the square of a number
- Find the cube of a number

Call these methods without creating any object.

```

class Calculator {

    static int square(int num) {
        return num * num;
    }

    static int cube(int num) {
        return num * num * num;
    }
}

public class Main {
    public static void main(String[] args) {

        int number = 5;
        System.out.println("Number = " + number);
        System.out.println("Square = " + Calculator.square(number));
        System.out.println("Cube = " + Calculator.cube(number));
    }
}

```

Number = 5

Square = 25

Cube = 125

Experiment 4: Static Block Initialization

Create a class **College** where:

- College Name
- University Name

are initialized inside a static block.

Display the values from the main() method and observe the execution order.

```
class College {  
  
    static String collegeName;  
    static String universityName;  
  
    static {  
        System.out.println("Static block is executed.");  
  
        collegeName = "University VOC College of Engineering";  
        universityName = "Anna University";  
    }  
  
    public static void main(String[] args) {  
  
        System.out.println("Main method is executed.");  
  
        System.out.println("College Name   : " + College.collegeName);  
        System.out.println("University Name : " + College.universityName);  
    }  
}
```

Static block is executed.

Main method is executed.

College Name : University VOC College of Engineering

University Name : Anna University

Experiment 5: Multiple Static Blocks

Write a Java program containing:

- Three static blocks
- One constructor
- One main method

Display messages from each and identify the execution sequence.

```
class Demo {  
  
    static {  
        System.out.println("Static Block 1 is executed.");  
    }  
    static {  
        System.out.println("Static Block 2 is executed.");  
    }  
  
    static {  
        System.out.println("Static Block 3 is executed.");  
    }  
  
    Demo() {  
  
    }  
}
```

```

        System.out.println("Constructor is executed.");
    }

    public static void main(String[] args) {

        System.out.println("Main method is executed.");
        Demo obj = new Demo();
    }
}

```

Static Block 1 is executed.
 Static Block 2 is executed.
 Static Block 3 is executed.
 Main method is executed.
 Constructor is executed.

```

class Demo {

    static {
        System.out.println("Static Block 1 is executed.");
    }
    static {
        System.out.println("Static Block 2 is executed.");
    }

    static {
        System.out.println("Static Block 3 is executed.");
    }

    Demo() {
        System.out.println("Constructor is executed.");
    }
}

public class Main{
    public static void main(String[] args) {

        System.out.println("Main method is executed.");
        Demo obj = new Demo();
    }
}

```

Main method is executed.
 Static Block 1 is executed.
 Static Block 2 is executed.
 Static Block 3 is executed.
 Constructor is executed.

Experiment 6: Static Nested Class

Create an outer class University containing:

- Static variable universityName

Create a static nested class Department that displays the university name and department name.

```
class University {  
  
    static String universityName = "Anna University";  
  
    static class Department {  
  
        String departmentName;  
  
        Department(String departmentName) {  
            this.departmentName = departmentName;  
        }  
  
        void display() {  
            System.out.println("University Name : " + University.universityName);  
            System.out.println("Department Name : " + departmentName);  
        }  
    }  
}  
  
public class Main {  
    public static void main(String[] args) {  
  
        University.Department d1 = new University.Department("Computer Science and  
Engineering");  
        University.Department d2 = new University.Department("Information Technology");  
  
        d1.display();  
        System.out.println();  
  
        d2.display();  
    }  
}
```

University Name : Anna University
Department Name : Computer Science and Engineering

University Name : Anna University
Department Name : Information Technology